

Research Summary Statistics			
Dataset	Records	Key Metrics	Time Period
M-Lab NDT (US)	50,000	RTT, Throughput, Loss	Oct-Nov 2025
Lumos5G (Mobile)	52,374	RSRP, RSRQ, SINR, Throughput	2020
Metric	Correlation (r)	R² Explained	Significance
RTT vs Throughput	-0.161	2.58%	p < 0.001
Loss vs Throughput	-0.111	1.23%	p < 0.001
RSRP vs Throughput (5G)	0.473	22.3%	p < 0.001
Model	R² (Test)	Median Throughput	vs RTT-Only
RTT-Only Selection	N/A	142.9 Mbps	Baseline
Linear Regression	5.7%	123.3 Mbps	-13.7%
Ridge Regression	11.7%	122.3 Mbps	-14.4%
Neural Network	20.5%	140.3 Mbps	-1.8%